Experiment: 06

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Subject Name: Competitive Coding-III **Subject Code: 23CSP-403**

1. **Aim:** Write a Program to Implement merge sort using the divide and Conquer Technique for recursive data sorting

2. **Objective:**

- Understand the **Divide and Conquer** paradigm: Divide, Conquer (Recursion), and Combine (Merge), and relate each phase to Merge Sort.
- Implement **Merge Sort recursively** and trace how an array of size *n* is divided into smaller subarrays across **log n levels**.
- Implement the **Merge** subroutine using two pointers to combine two sorted halves in **O(n) time**.
- Count **comparisons and recursive calls** to verify the **O(n log n)** time complexity of Merge Sort.
- Analyse the **auxiliary space complexity**, including **O(n)** temporary arrays and **O(log n)** recursion stack space.
- Apply **Merge Sort and its merge-step logic** to solve a hard LeetCode problem and understand its guaranteed **O(n log n)** performance for large inputs.

3. **Software**: VS Code, LeetCode

4. **Code**

```
import heapq

class Solution:
    def mergeKLists(self, lists):
        heap = []

        for i in range(len(lists)):
            if lists[i]:
                heapq.heappush(heap, (lists[i].val, i, lists[i]))

        dummy = ListNode(0)
        curr = dummy

        while heap:
            val, i, node = heapq.heappop(heap)
```

```
            curr.next = node
            curr = curr.next

            if node.next:
                heapq.heappush(heap, (node.next.val, i, node.next))

        return dummy.next
```

5. Output

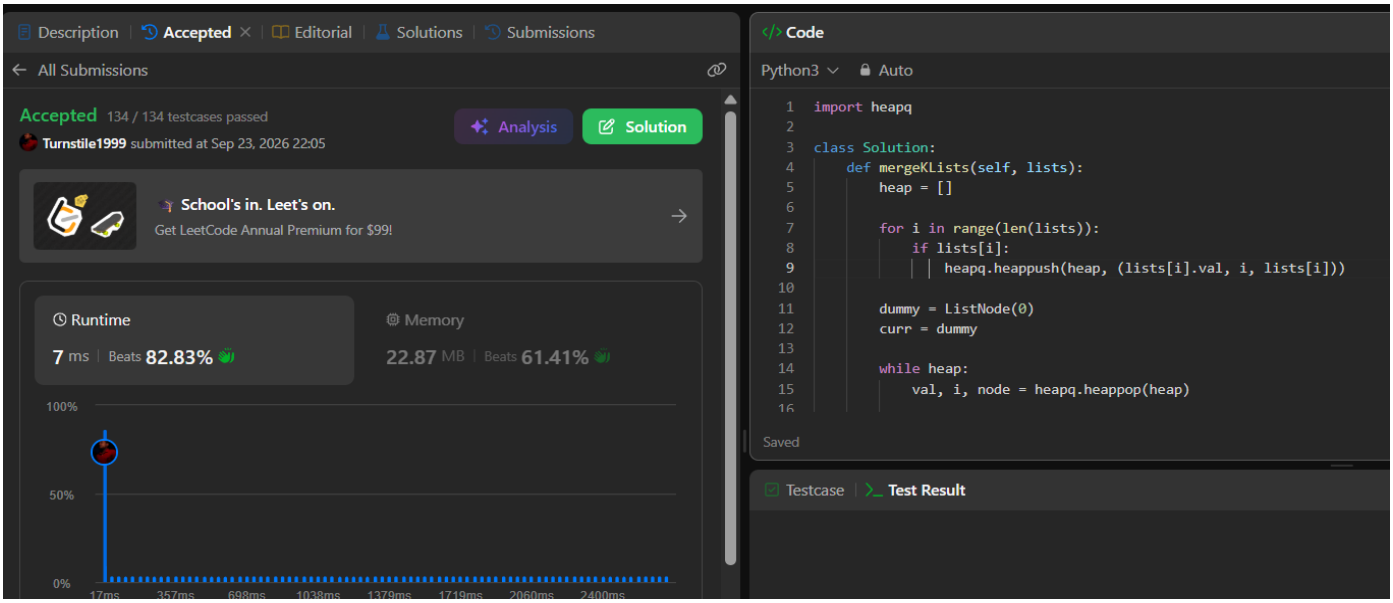


Fig 1

7. Learning Outcomes:

- Understand the concept of **merging k sorted linked lists** using a Min Heap.
- Implement a **Min Heap** to efficiently select the smallest node from multiple sorted linked lists.
- Apply **priority queue operations** such as insertion and deletion to maintain the smallest available element.
- Analyse the **time and space complexity** of the heap-based approach as **O(N log k)** and **O(k)** respectively.
- Apply the **Min Heap technique** to solve **LeetCode 23 – Merge k Sorted Lists** efficiently.